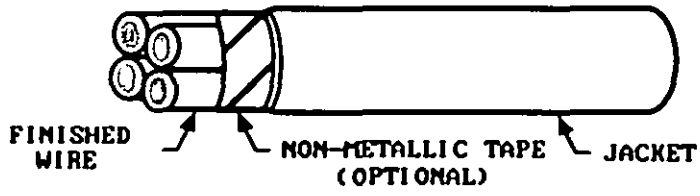
INCH-POUND
MIL-C-24640/3B
31 July 1989
SUPERSEDING
MIL-C-0024640/3A(SH)
23 October 1985
USED IN LIEU OF
MIL-C-24640/3
10 August 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT, FOUR CONDUCTOR, POWER, TYPE FX (INCLUDING VARIATION TYPE FXA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.



SH 12750

FINISHED WIRE:

(B) MIL-W-81044/12 except identification codes and methods shall be in accordance with MIL-C-24640. Standard identification code applied by method 1 or 3.

FIGURE 1. Construction (non-watertight).

(B) denotes changes.

AMSC N/A FSC 6145

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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(B) TABLE I. Construction details-unarmored.

PIN M24640/3	Type number	AWG size	Conductor diameter (inch) (nominal)	Finished wire diameter (inch) (min)	Finished wire diameter (inch) (max)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01UN	FX-3	16	0.055	0.069	0.075	0.034	0.262	0.282	63.0	4.98
-02UN	FX-4	14	.069	.085	.093	.038	.311	.335	92.6	3.18

(B) TABLE II. Construction details - armored.

PIN M24640/3	Type number	AWG size	Conductor diameter (inch) (nominal)	Finished wire diameter (inch) (min)	Finished wire diameter (inch) (max)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01AN	FXA-3	16	0.055	0.069	0.075	0.034	0.312	0.332	80.0	4.98
-02AN	FXA-4	14	.069	.085	.093	.038	.361	.385	112.6	3.18

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REQUIREMENTS:

Qualification required.
Construction details: See tables I and II.

When this cable is manufactured with armor, the following additional requirements shall apply:

(B) (a) Braided metal armor shall be added.
(b) Cable surface marking will not be required.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	See tables I and II
(B) Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	Not applicable
Shield to shield	Not applicable
Insulation resistance - megohms/1000 feet, minimum	5000
Conductor and shield continuity	Not applicable
Jacket flaws	Not applicable
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
Capacitance unbalance - percent, maximum	Not applicable
Characteristic impedance - at 1 MHz, ohms	Not applicable
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
Weight	No failure
Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
Gas flame (1 hour)	Not applicable

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EXAMINATION AND TESTS: - Continued

	Requirements
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor (type FXA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	Not applicable
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	Not applicable
Surface transfer impedance	
Milliohms per meter, maximum	Not applicable
EMP response, dB, minimum	Not applicable
Pulse response time - ms/500 feet	
Rise time, maximum	Not applicable
Decay time, maximum	Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50

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QUALIFICATION INSPECTION: - Continued

	Requirements
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Finished wire	Not applicable

Preparing activity:
Navy - SH
(Project 6145-1004-03)

Custodians:
Army - MI
Navy - SH
Air Force - 85

Review activities:
Navy - EC
Air Force - 99
DLA - IS

User activities:
Army - ME, AR
Navy - CG

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